Summary

	Bit Rate =	Coding Rate ×	Baud Rate
	bps	$\frac{\#\text{ bits}}{\text{pulse}}$	$\frac{\#\text{pulses}}{\text{sec}}$
Channel Bandwidth	✓		✓
Pulse Shape	✓	✓	✓
SNR	✓	✓	

max bit rate
= Shannon channel capacity
(for AWGN channel)

ISI
distortion

Nyquist Rate
(max Baud rate)

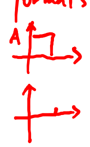
Line Coding

① Unipolar NRZ (Non Return to Zero)

- 2 signal formats

+A : "1"

0 : "0"



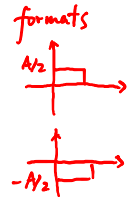
* poor timing recovery

② Polar NRZ

- 2 signal formats

$+\frac{A}{2}$

$-\frac{A}{2}$

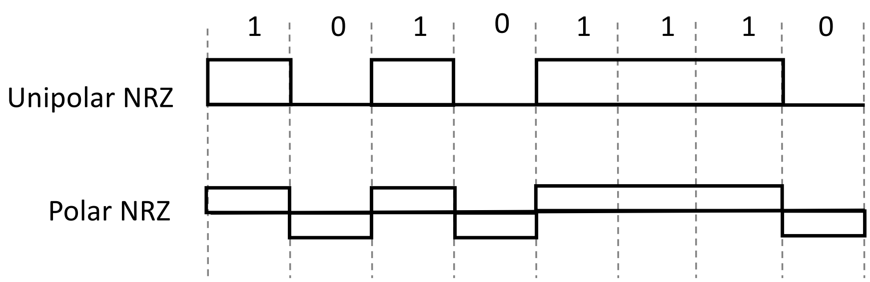


What is Line Coding?

- One method to convert a binary information sequence into signals that enter the communication channel
  - E.g., “1” maps to +A/2 square pulse; “0” to –A/2 square pulse polar NRZ

- Design considerations:
  - Timing recovery *
  - Low complexity and implementation cost
  - Low power and energy efficient
  - Better immunity to noise and interference
  - Built-in error detecting capability

Unipolar & Polar Non-Return-to-Zero (NRZ) Coding

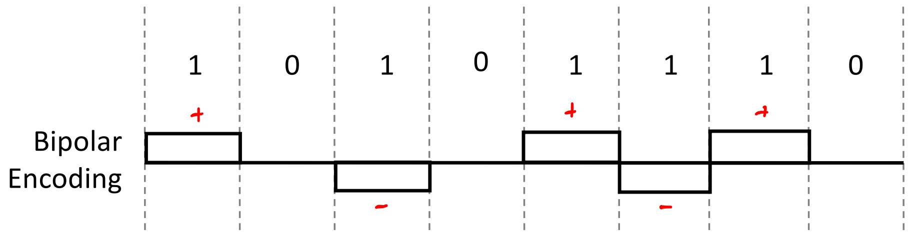


Unipolar NRZ
- “1” maps to +A pulse
- “0” maps to no pulse
- Average Power: High
  $0.5*A^2 + 0.5*0^2 = A^2/2$
- Long string of “1”s or “0”s
  - Poor timing
- Simple

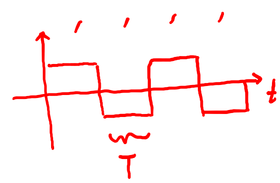
Polar NRZ
- “1” maps to +A/2 pulse
- “0” maps to –A/2 pulse
- Average Power: Lower
  $0.5*(A/2)^2 + 0.5*(-A/2)^2 = A^2/4$
- Long string of “1”s or “0”s
  - Poor timing
- Simple

3

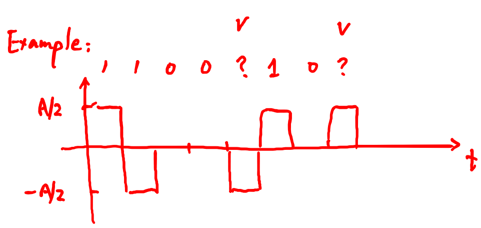
Bipolar Coding



- Three signal formats: {–A/2, 0, +A/2}
- "1" maps to +A/2 or –A/2 in alternation
- "0" maps to no pulse
  - Every + pulse matched by – pulse
- String of "1"s produces a square wave
  - Spectrum centered at 1/(2T)
- Long string of "0"s causes receiver to lose synch
- Zero-substitution codes *



Example:



Bipolar violation

- error detection
- used for "zero substitution"

0 0 0 0 0 0 0 0

⇓

0 0 0 V 1 0 V 1

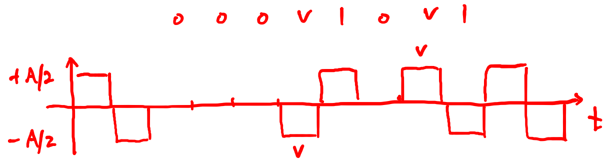
B8ZS

Bipolar with
8
Zeros
Substitution

Example:

1 1 "0 0 0 0 0 0 0 0" 1 1

0 0 0 V 1 0 V 1



used in
North American
T1 system
(1.544 Mbps)

B6ZS

0 0 0 0 0 0

↓

0 V 1 0 V 1

used in
T2 system
(6.312 Mbps)

B3ZS

0 0 0

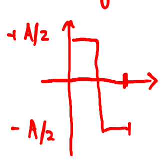
↓

0 0 V or 1 0 V

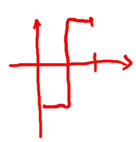
used in
T3 system
(44.716 Mbps)

④ Manchester Coding (RZ)

– 2 signal formats

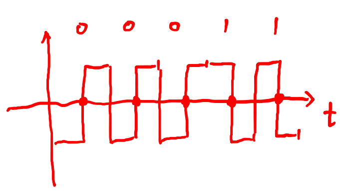


"1"



"0"

Example:



- good timing recovery
- "self-clocking" at RZ points

$$\text{Coding rate} = \frac{1 \text{ bit}}{2 \text{ pulses}} = \frac{1}{2} \text{ bit/pulse}$$

bit rate = coding rate × Baud rate

E.g. 10 Mbps = $\frac{1}{2}$ × ? 20 MHz

higher Baud rate

Application : used in
bandwidth-rich system
such as Ethernet

Also known as

$\underbrace{1B}_{\text{1 Binary bit}}\underbrace{2B}_{\text{2 Binary pulses}}$ coding   coding rate $= \frac{1}{2}$

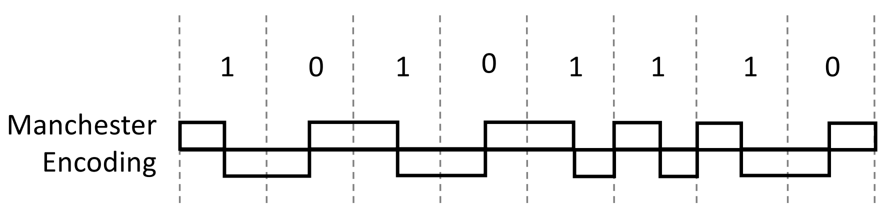
$\Rightarrow$ part of $\underbrace{m B}_{\text{m Binary bits}}\underbrace{n B}_{\text{n Binary pulses}}$ family   coding rate $= \frac{m}{\underbrace{n}} \leq 1$ for $mBnB$

$\underline{2^m \leq 2^n}$ $\Rightarrow$ $m \leq n$

E.g. 4B5B $\Rightarrow$ 100 Mbps Ethernet

64B66B $\Rightarrow$ 10 Gbps Ethernet

Manchester Coding & *m*B*n*B Coding



Manchester

- “1” maps to A/2 first T/2, -A/2 last T/2
- “0” maps to –A/2 first T/2, A/2 last T/2
- Every interval has transition in middle
  - Easy timing recovery
  - Double the minimum bandwidth
- Simple to implement
- Used in 10 Mbps Ethernet & other LAN systems

*m*B*n*B

- Maps block of *m* information bits into *n* pulses
- Manchester code is *1*B*2*B code
- *4*B*5*B code is used in 100 Mbps Ethernet
- *8*B*10*B code is used in Gigabit Ethernet
- *64*B*66*B code is used in 10 Gbps Ethernet

1B2B in 10 Gbps Ethernet ?